

Project Overview

This project aims to create a basic serial calculator using an Arduino Uno. This allows users to perform simple arithmetic operations (addition, subtraction, multiplication, division) by entering inputs through a PC console application. This acts as the front-end to collect user input and display the results and the Arduino processes the input string and returns the result over serial communication.

This offloads processing from the PC to the microcontroller and mimics a basic client-server model over a serial connection.

Objectives

1. Develop a PC console application written in C++ for user interaction.
2. Develop a serial communication link over COM port between the Arduino Uno and the console app.
3. Accept and parse user input in real-time from console application on PC to Arduino over a serial port.
4. Calculation of input on the microcontroller.
5. Display results back on the console application.
6. Handles error for invalid inputs such as wrong syntax.

Functional Requirements

1. The system must be able to receive input in the form: number1 Operator number2.
2. Supports the basic operators: +, -, *, /
3. It must return a result or an error message back to the user.

Tools and Components

- Microcontroller: Arduino Uno
- USB cable for serial communication
- User interface: Console application
- Language: C++ (Arduino development), C++ (Console development)
- Serial library: ManashMandal SerialPort library.
- Development Environment:
 - Arduino IDE for the sketch
 - Visual Studio for the PC application

Why C++?

- C++ provides low-level access to the serial port using a serial port library.
- Similarity of the Console application with Arduino code which is C/C++ allows the sharing of logic between both applications.

Reference

Mandal, M (2020, April 21). *SerialPort*. Github. <https://github.com/manashmandal/SerialPort>